

## EILDON FIRE TOWER, Australia

WMO: 948810

Lat: 37.209S

Lon: 145.842E

Elev: 637

StdP: 93.90

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 1.5        | 2.3 | -2.2                          | 3.4 | 13.5 | -1.0 | 3.8 | 9.7  | 13.8                  | 5.6  | 12.1 | 5.9  | 5.0                   | 260  | 0.603 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 12.6                   | 32.6            | 17.3 | 30.4 | 16.8 | 28.6 | 16.3 | 19.5                | 26.5 | 18.6 | 26.0 | 17.8 | 25.3 | 4.9                  | 50   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.7                            | 13.7 | 20.9 | 16.3 | 12.5 | 20.5 | 15.2 | 11.6 | 20.0 | 58.3          | 26.4 | 55.1 | 25.8 | 52.6 | 25.5 | 23.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 12.0              | 10.4 | 9.3 | DB                         | 0.1  | 36.4               | 0.6 | 1.9                                                | -0.3 | 37.7       | -0.7 | 38.8       | -1.0 | 39.9       | -1.4 | 41.3 |
|                   |      |     | WB                         | -0.5 | 20.9               | 0.5 | 1.2                                                | -0.8 | 21.8       | -1.2 | 22.5       | -1.5 | 23.2       | -1.9 | 24.1 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.8        | 20.0  | 19.7  | 17.2  | 13.4  | 9.7   | 7.3   | 6.4   | 7.0   | 9.2   | 11.8  | 14.9  | 17.2  | (6)  |
|                                                                        | DBStd       | 5.90        | 4.62  | 4.27  | 4.18  | 3.42  | 2.65  | 2.00  | 1.80  | 2.20  | 2.80  | 3.70  | 4.39  | 4.52  | (7)  |
|                                                                        | HDD10.0     | 416         | 0     | 0     | 1     | 9     | 38    | 84    | 112   | 96    | 47    | 21    | 6     | 2     | (8)  |
|                                                                        | HDD18.3     | 2288        | 35    | 33    | 72    | 150   | 267   | 331   | 369   | 351   | 273   | 206   | 123   | 78    | (9)  |
|                                                                        | CDD10.0     | 1430        | 309   | 271   | 224   | 111   | 29    | 3     | 1     | 4     | 24    | 77    | 153   | 224   | (10) |
|                                                                        | CDD18.3     | 261         | 86    | 70    | 38    | 3     | 0     | 0     | 0     | 0     | 0     | 3     | 19    | 42    | (11) |
|                                                                        | CDH23.3     | 2563        | 959   | 656   | 302   | 11    | 0     | 0     | 0     | 0     | 0     | 22    | 170   | 444   | (12) |
|                                                                        | CDH26.7     | 898         | 396   | 228   | 82    | 1     | 0     | 0     | 0     | 0     | 0     | 2     | 40    | 151   | (13) |
| (14) Wind                                                              | WSAvg       | 4.7         | 5.1   | 4.7   | 4.6   | 4.3   | 4.3   | 4.3   | 4.6   | 4.8   | 5.1   | 5.0   | 5.0   | 5.0   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 791         | 47    | 51    | 47    | 54    | 62    | 96    | 89    | 86    | 80    | 66    | 62    | 49    | (15) |
|                                                                        | PrecMax     | 1259        | 124   | 156   | 133   | 111   | 135   | 173   | 166   | 161   | 162   | 142   | 137   | 123   | (16) |
|                                                                        | PrecMin     | 392         | 0     | 0     | 13    | 14    | 1     | 33    | 24    | 19    | 17    | 8     | 18    | 2     | (17) |
|                                                                        | PrecStd     | 183         | 37    | 42    | 28    | 29    | 35    | 37    | 35    | 42    | 39    | 34    | 31    | 32    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 36.8  | 34.5  | 32.0  | 24.8  | 18.6  | 14.1  | 12.7  | 15.6  | 20.5  | 26.3  | 30.2  | 34.1  | (19) |
|                                                                        |             | MCWB        | 17.7  | 18.3  | 16.3  | 14.1  | 12.1  | 10.5  | 9.2   | 10.8  | 12.4  | 14.5  | 16.0  | 16.8  | (20) |
|                                                                        | 2%          | DB          | 33.3  | 31.4  | 28.9  | 22.4  | 16.3  | 12.5  | 11.0  | 13.5  | 17.6  | 23.1  | 27.5  | 30.6  | (21) |
|                                                                        |             | MCWB        | 17.8  | 17.4  | 16.1  | 13.8  | 11.6  | 9.6   | 8.5   | 9.8   | 11.9  | 13.6  | 15.6  | 16.0  | (22) |
|                                                                        | 5%          | DB          | 30.6  | 29.4  | 26.6  | 20.7  | 14.9  | 11.4  | 10.0  | 11.9  | 15.8  | 20.8  | 25.1  | 27.9  | (23) |
|                                                                        |             | MCWB        | 17.0  | 17.0  | 15.6  | 13.2  | 11.2  | 9.0   | 7.9   | 8.7   | 11.2  | 12.8  | 14.8  | 15.6  | (24) |
|                                                                        | 10%         | DB          | 28.5  | 27.5  | 24.5  | 19.1  | 13.7  | 10.3  | 9.1   | 10.6  | 14.1  | 18.5  | 22.8  | 25.5  | (25) |
|                                                                        |             | MCWB        | 16.5  | 16.4  | 15.1  | 12.6  | 10.4  | 8.3   | 7.4   | 8.0   | 10.2  | 12.3  | 14.1  | 15.0  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 20.8  | 21.1  | 19.2  | 16.4  | 14.0  | 11.6  | 10.3  | 11.7  | 14.3  | 16.7  | 18.8  | 19.2  | (27) |
|                                                                        |             | MCDB        | 27.5  | 27.0  | 25.4  | 21.2  | 15.7  | 12.7  | 11.2  | 14.3  | 17.2  | 22.6  | 24.9  | 25.6  | (28) |
|                                                                        | 2%          | WB          | 19.6  | 19.4  | 17.8  | 14.9  | 12.6  | 10.3  | 9.2   | 10.2  | 12.9  | 14.9  | 17.1  | 17.9  | (29) |
|                                                                        |             | MCDB        | 27.2  | 26.2  | 24.2  | 19.9  | 14.9  | 11.7  | 10.3  | 12.6  | 16.4  | 19.7  | 23.8  | 24.8  | (30) |
|                                                                        | 5%          | WB          | 18.6  | 18.5  | 16.9  | 14.0  | 11.7  | 9.3   | 8.3   | 9.1   | 11.6  | 13.9  | 16.1  | 16.8  | (31) |
|                                                                        |             | MCDB        | 27.1  | 25.6  | 23.6  | 18.5  | 14.2  | 10.7  | 9.5   | 11.3  | 15.1  | 18.5  | 22.7  | 24.6  | (32) |
|                                                                        | 10%         | WB          | 17.6  | 17.7  | 16.0  | 13.3  | 10.8  | 8.6   | 7.6   | 8.2   | 10.6  | 12.9  | 15.1  | 15.9  | (33) |
|                                                                        |             | MCDB        | 26.3  | 24.9  | 22.4  | 17.9  | 13.3  | 9.9   | 8.8   | 10.3  | 13.8  | 17.6  | 21.1  | 23.6  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 12.6  | 11.8  | 10.3  | 8.1   | 5.5   | 4.5   | 4.3   | 5.3   | 7.0   | 9.0   | 10.7  | 11.8  | (35) |
|                                                                        |             | MCDBR       | 14.7  | 13.7  | 12.5  | 10.1  | 7.2   | 5.9   | 5.5   | 7.2   | 9.1   | 11.7  | 13.1  | 14.4  | (36) |
|                                                                        | 5% WB       | MCWBR       | 4.8   | 4.4   | 4.3   | 4.5   | 4.2   | 4.2   | 3.8   | 4.7   | 5.4   | 5.9   | 5.3   | 4.8   | (37) |
|                                                                        |             | MCWBR       | 12.8  | 11.8  | 10.4  | 8.7   | 6.2   | 4.9   | 4.7   | 6.4   | 8.1   | 10.1  | 11.3  | 11.9  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.324 | 0.325 | 0.304 | 0.297 | 0.283 | 0.275 | 0.273 | 0.282 | 0.306 | 0.308 | 0.319 | 0.319 | (40) |
|                                                                        |             | taud        | 2.548 | 2.553 | 2.613 | 2.597 | 2.619 | 2.635 | 2.642 | 2.605 | 2.522 | 2.548 | 2.543 | 2.555 | (41) |
|                                                                        | Ebn at Noon |             | 1010  | 986   | 970   | 917   | 875   | 855   | 878   | 921   | 951   | 993   | 1008  | 1020  | (42) |
|                                                                        |             | Edh at Noon | 108   | 103   | 89    | 79    | 67    | 61    | 65    | 77    | 96    | 102   | 108   | 108   | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 7.30  | 6.37  | 5.18  | 3.63  | 2.41  | 1.84  | 2.02  | 2.86  | 4.08  | 5.48  | 6.46  | 7.21  | (44) |
|                                                                        | RadStd      |             | 0.45  | 0.39  | 0.29  | 0.22  | 0.17  | 0.20  | 0.12  | 0.23  | 0.36  | 0.49  | 0.38  | 0.39  | (45) |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (41 neighbors) |       | +0.50   | N/A    | N/A     | +0.88 | N/A   | N/A         | -18     | -125    | +174    |

Nomenclature: See separate page